NOVA AI STUDY ASSISTANT

Comprehensive Technical Architecture & Tech Stack Manual

Generated on: October 2026 • Production Edition • Full-Stack Web App

1. EXECUTIVE SUMMARY

Nova AI Study Assistant is an enterprise-grade academic learning companion designed for university students (featuring official AKTU CBCS curriculums for B.Tech CSE). The system integrates intelligent generative AI tutoring, syllabus unit tracking, interactive countdown tests, active recall flashcards, and timetable management into a fast, offline-resilient full-stack web application with zero third-party framework dependencies in the presentation layer.

2. CORE SYSTEM SPECIFICATIONS

Runtime Platform	Node.js (LTS v18/v20) + Express.js 4.19
Frontend Paradigm	Modern Semantic HTML5, CSS3 Custom Properties, Vanilla ES6+
Artificial Intelligence	Google Gemini 2.5 Flash via @google/genai SDK (v2.21.0)
Database Architecture	Tiered: MongoDB Atlas (Mongoose v8.5) + Local JSON Store + Client IndexedDB
Cloud & Integration	Firebase Firestore / Auth SDK (v13.0.0) + RESTful Microservices
Active Port & Binding	Port 3000 (0.0.0.0 default with CORS enablement)
Offline Capability	Browser-side IndexedDB persistence for user tasks, flashcard decks & sessions

3. ARCHITECTURAL HIGHLIGHTS

Zero-Framework Frontend

Built using pure ES6+ JavaScript modules. Instant cold-start rendering, zero bundle bloat, zero build latency, and high accessibility compliance.

Triple-Layer Resilience

Transparent fallback pipeline: MongoDB Atlas -> Server-side JSON files -> Browser IndexedDB. Data never drops even during network partitions.

Server-Side Gemini API

Direct server-side AI proxy via @google/genai keeps client API keys secure, enables context priming, and streams answers safely to students.

AKTU CBCS Knowledge Base

Pre-seeded syllabus registries for BCS-301, BCS-302, BAS-303, BAS-301, BCS-303, and BCC-302 with verified PYQ question banks.

4. LAYER-BY-LAYER TECHNICAL BREAKDOWN

• Vanilla JavaScript Architecture (miniproject/app.js, timetable.js, tests-planner.js, courses.js): Modularity without node_modules bundles.

• Responsive HTML5 Single Page Application (SPA): Dynamic view switcher toggling Dashboard, AI Chat, Flashcards, Calendar, Courses, Assignments, and Tests.

• Design System (style.css, timetable.css, tests-planner.css): Academic Midnight & Slate aesthetic, Outfit & Plus Jakarta Sans typography, JetBrains Mono for code blocks.

• Floating Interactive AI Bubble (miniproject/nova-fab.js): Persistent bottom-right floating action widget allowing spontaneous question querying anywhere in app.

• Local Client Database (miniproject/db.js): Full IndexedDB wrapper with object stores for users, decks, tasks, notes, and local session caching.

• Node.js & Express.js (v4.19.2): High-throughput non-blocking asynchronous REST API server handling student sessions and coursework.

• Payload Scaling: Extended JSON and urlencoded body parsers up to 25MB allowing in-depth lecture PDF note text and image submission.

• Specialized Microservices (backend/tests-exam-service.js & timetable-service.js): Isolated business logic for grading, analytics, and section scheduling.

• Dual Database Abstraction: Dynamically detects whether MongoDB Atlas is reachable; automatically redirects writes to thread-safe local JSON persistence if offline.

• Security: Client-side secrets are isolated server-side. GEMINI_API_KEY is read through process.env and never exposed to the frontend.

• SDK: Official modern Google Gen AI TypeScript/JavaScript SDK (@google/genai v2.21.0) utilizing gemini-2.5-flash for real-time inference.

• Pedagogical System Instruction: Embedded prompts instructing Gemini to act as a university computer science and engineering mentor.

• Multi-Conversation Memory: Persistent conversation trees allowing students to maintain subject-specific chat sessions without losing context.

• Syllabus Grounding: Injects AKTU course syllabi (BCS-301 Data Structures, BCS-302 COA, etc.) to tailor answers to semester exams.

• PDF Summarization & Quiz Extraction: Generates structured chapter summaries, flashcards, and high-yield questions on command.

• Primary Cloud Store: MongoDB Atlas via Mongoose 8.5 with schemas for Chats, Conversations, User Profiles, Tasks, and

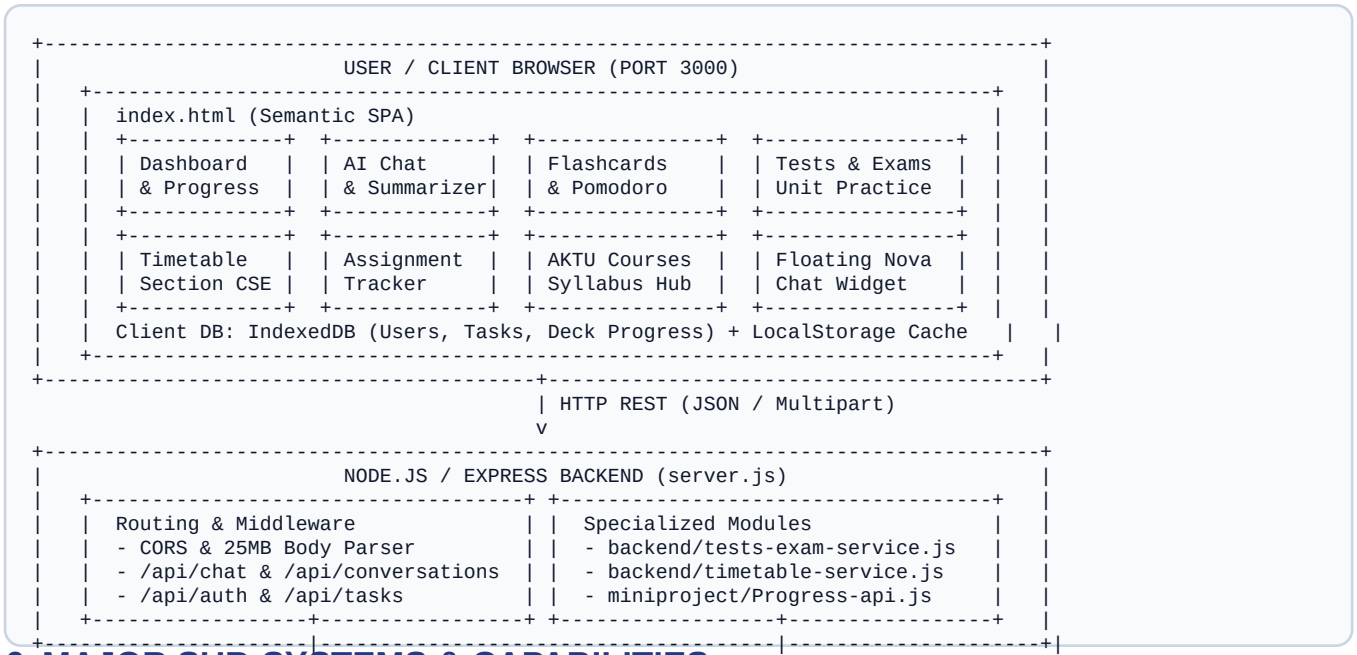
Assignments.

• Zero-Config Local Store (/data/*.json): chat-store.json, user-store.json, tasks-store.json, assignments-store.json, timetable-store.json.

• Client-Side Storage: IndexedDB (users, decks, tasks, flashcard progress) + LocalStorage (auth tokens, exam countdown states, UI filters).

• Firebase Integration: Configured through firebase-applet-config.json and firebase v13 for persistent identity and cloud Firestore synchronization.

5. SYSTEM ARCHITECTURE & DATA FLOW DIAGRAM



6. MAJOR SUB-SYSTEMS & CAPABILITIES

- **AI Study Chat & Multimodal Summarization:** Interactive conversational tutor grounded in student course memory. Includes file attachment support for summarizing PDF lecture notes and generating multi-choice practice drills.
- **Official AKTU CBCS Courseware Registry:** Unit-by-unit syllabus outlines, weightage allocation, prescribed textbook lists, and university exam codes for BCS-301 (DS), BCS-302 (COA), BAS-303 (Maths IV), and others.
- **Tests & PYQ Assessment Engine:** Timed online exam simulation with questions palette, answer review flags, instant grading, 6-metric scorecard, and automated unit-by-unit syllabus mastery percentage bars.
- **Interactive Timetable Service:** Section-based schedule management (CSE 2A through CSE 2D) with real-time class status detection (Active Now, Next Class, Break), room assignments, and professor mappings.
- **Active Recall Flashcards & Pomodoro Timer:** Curriculum-aligned flashcard decks with spaced-repetition feedback (Again, Hard, Good, Easy) paired with a 25-minute Pomodoro focus interval system.
- **Assignment Tracker & Academic Deadlines:** Comprehensive coursework ledger with priority sorting (High/Med/Low), due-date indicators (Overdue, Due Soon in 72h, Completed), and status notifications.

7. REST API ENDPOINT SPECIFICATION

POST	`/api/chat`	Send message to Gemini AI, returns answer and persists chat history
GET	`/api/conversations`	Retrieve all saved conversation threads for authenticated student
POST	`/api/conversations`	Create a new independent conversational thread with academic context
POST	`/api/login / /signup`	Authenticate user against MongoDB or persistent local credential store
GET / POST	`/api/tasks`	Query or append daily study tasks and goal milestones
GET / POST	`/api/assignments`	CRUD endpoints for tracking course assignments, due dates and links
GET	`/api/timetable/:section`	Fetch weekly lecture and lab timetable for chosen CSE section
POST	`/api/tests/evaluate`	Evaluate submitted mock or unit test; computes score & unit breakdowns
GET	`/api/tests/analytics/:email`	Calculate aggregate statistics, average percentage, and mastery rates
GET	`/api/tech-stack-pdf`	Download / stream this comprehensive technical architecture manual

8. SECURITY, PERFORMANCE & PORTABILITY

• **API Key Isolation & Proxying:** The Gemini API token is strictly scoped to the server runtime. Frontend requests never interface with third-party AI providers directly, mitigating token extraction risks.

• **Zero Build Step Portability:** The entire application runs directly on standard Node.js without requiring webpack, vite, or babel transpilations. npm start boots the complete system within 2 seconds.

• **Input Sanitization & Protection:** Cross-Site Scripting (XSS) defenses are built into escapeHtml utilities in app.js, while CORS restrictions enforce cross-origin resource access safety.

• **Container & Cloud Ready:** Configured to run effortlessly in Docker, Google Cloud Run, Heroku, AWS EC2, or local school lab environments on port 3000.

Summary & Verification

Nova AI Study Assistant demonstrates how modern web standard technologies, paired with Google Gemini AI and resilient multi-tier persistence, deliver an ultra-responsive, zero-latency educational platform without the complexity and maintenance burden of bulky JavaScript frameworks.